



SMART INDIA HACKATHON 2025



Basic Details

- Problem Statement ID – **25032**
- Problem Statement Title- **Development of a Smart Digital Platform to Promote Eco & Cultural Tourism in Jharkhand**
- Theme- **Travel and Tourism**
- PS Category- **Software**
- Team ID - **76239**
- Team Name – **Bit-Storm**



Challenges and Problems

- **Lack of Digital Infrastructure** - No centralized platform for tourist information and services.
- **Limited Promotional Outreach** - Low visibility of Jharkhand's destinations at national and global levels.
- **Low Tourist Awareness** - Visitors struggle to find reliable info on destinations, routes, accommodations, and guides.
- **Unorganized Travel & Hospitality Services** - Fragmented transport, guide, and accommodation services.
- **Exclusion of Local Communities** - Tribal artisans, homestay providers, and small businesses are left out of the tourism value chain.
- **Trust & Safety Issues** - Lack of verified guides and secure transaction systems.
- **Underutilized Eco & Cultural Potential** - Natural beauty, tribal heritage, and cultural festivals remain untapped due to poor digital promotion.

Proposed Solution

➤ AI-Powered Itinerary Planner

Our solution is to develop an AI-powered itinerary planner that keeps **tourists' travel budget, goals, schedule and preferences** in mind to personalize a perfect travel plan for them. It also **suggests alternate plans** if you're willing to bend a few rules to get more out of the trip. The website and chatbot enable multi-lingual support. Interactive maps and AR/VR support allow the tourists to get a glimpse and to whet their anticipation.

➤ Digital cultural promotion

Along with itinerary planning, the platform provides a **digital marketplace for local tribal artisans**, homestay owners, and tour guides. It uses **blockchain-enabled certifications and transactions** to ensure fair trade, fewer scams, and an overall pleasant experience. The app also enables tourists and local traders to leave feedback.

➤ Data-driven Governance

AI-based feedback analysis allows tourism officials to track shortcomings and utilize ways and means to provide a peaceful and pleasant experience for locals and tourists alike. It also helps track and analyze trends, improve policies, and introduce new promotional events.

❖ Innovation and Uniqueness



❖ How it Addresses the Problems

Personalized Itinerary Planning: Custom travel plans based on interests, time, and budget.

Multilingual AI Chatbot: 24/7 support in multiple languages for instant query resolution.

Blockchain Trust System: Secure payments, verified providers, and digital certification for fair trade.

Interactive Real-Time Maps: Live public transport tracking, routes, and nearby attractions for easy navigation.

AR/VR Destination Previews: Virtual tours of sites and events to encourage in-person visits.

Integrated Digital Marketplace: Platform for selling local crafts, homestays, events, and eco-tourism packages.

Tourism Data Analytics: Tracks feedback, trends, weather, and festivals for informed policy and promotion.

Localized Data & Language: Uses local rainfall data and supports regional languages.

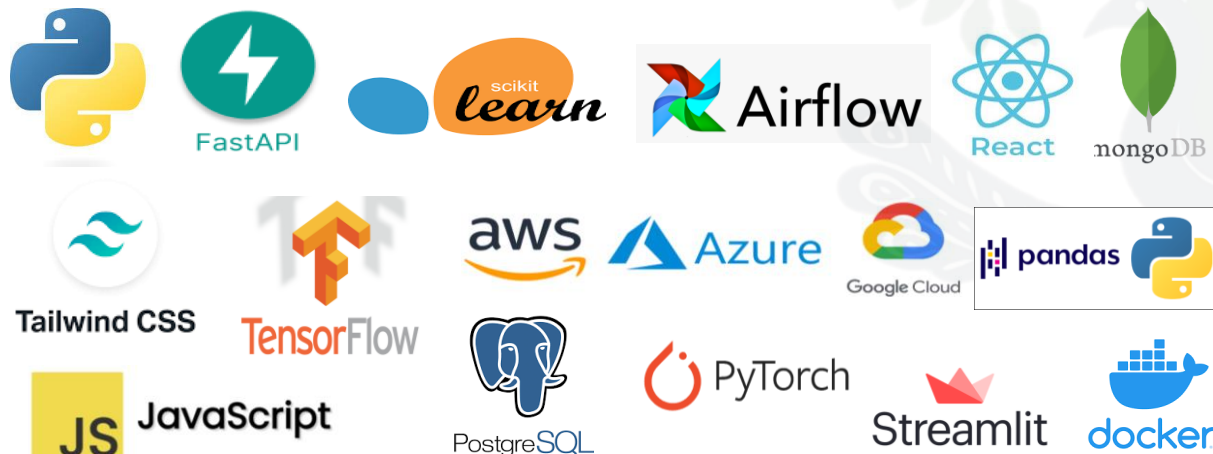
Civil Engineering Backing: Incorporates technical calculations (sizing, filters) with expert validation.

Future-Ready Features: Generates PDF reports and integrates government subsidies/marketplace links.

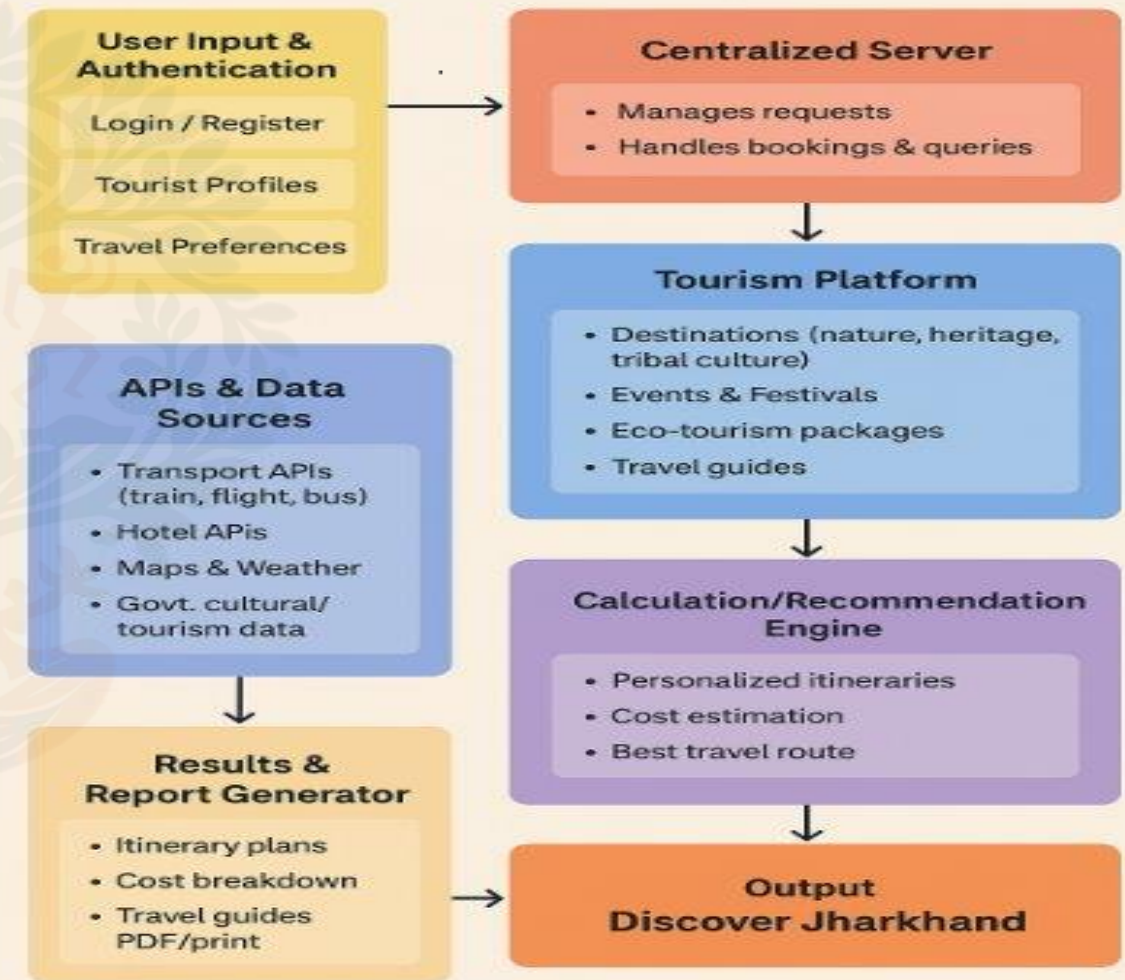
Methodology and Process

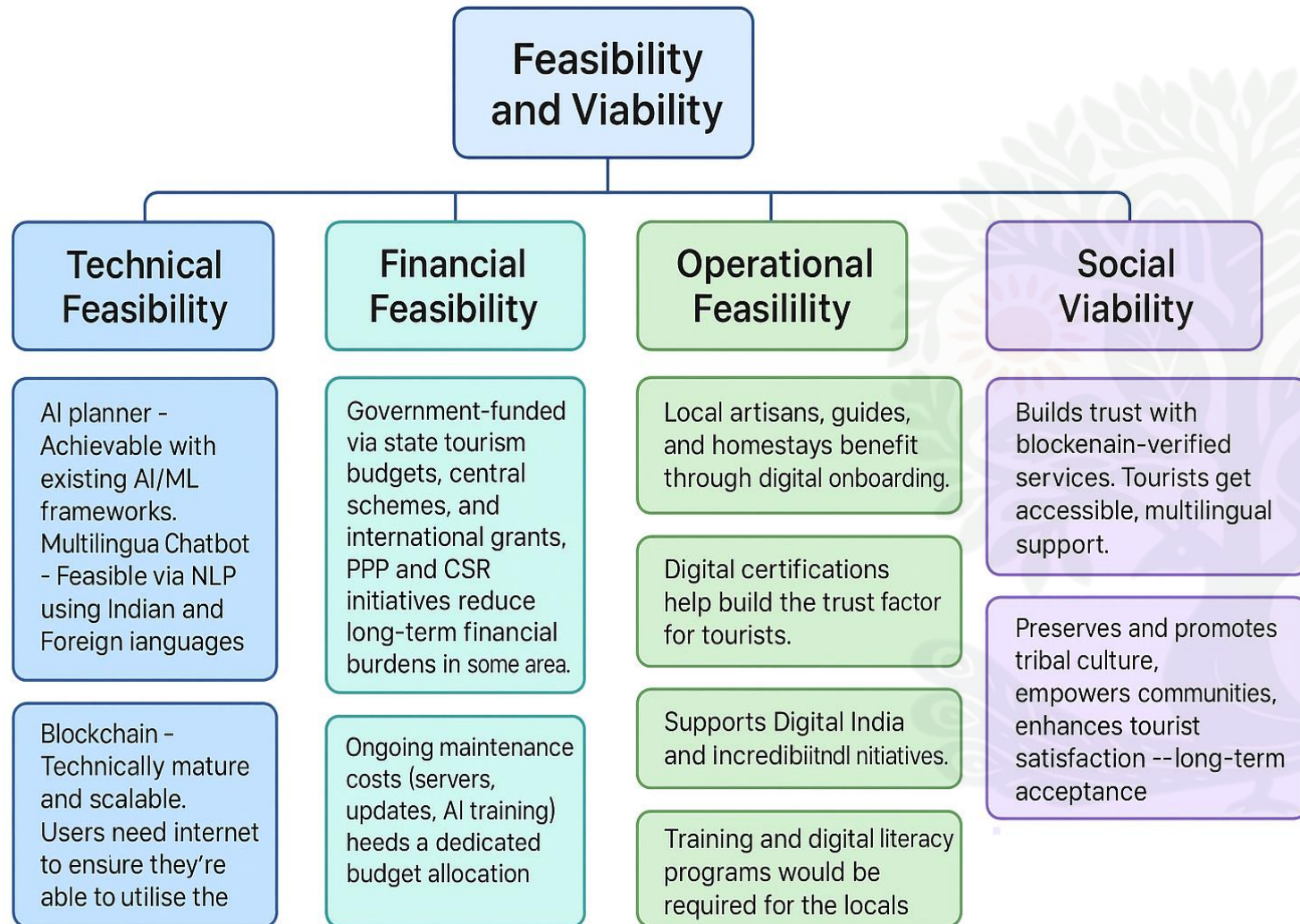
- **Data Collection & Integration:** Gather destination, transport, accommodation, cultural, and community data from government portals, IoT, **geo-location**, and user inputs.
- **Data Preprocessing & Storage:** Clean and standardize datasets; store securely on scalable cloud platforms.
- **AI/ML Model Development:** Build models for personalized itineraries, **sentiment analysis**, and recommendations.
- **Blockchain & Verification:** Verify guides/services, enable **secure payments**, and issue digital certificates.
- **Immersive & Real-Time Services:** Provide **AR/VR previews**, interactive maps, and live location/transport tracking.
- **Scaling & Improvement:** Expand coverage, integrate more communities, and retrain AI for better accuracy.

Technology Stack



Jharkhand Tourism Project





Strategies for Overcoming Challenges

TECHNICAL CHALLENGES

- **Offline Features:** Provide downloadable maps & itineraries for low-network zones.
- **Partnerships with Telecom Providers:** Collaborate with telecom providers to boost rural connectivity.
- **Lightweight App Design:** Optimized for low bandwidth and older smartphones.

FINANCIAL CHALLENGES

- **Government Schemes:** Central/state tourism development funds.
- **CSR & PPP Models:** Involve tech companies and telecoms as CSR partners.
- **Phased Rollout:** Initiate in key Tourist Hotspot districts before statewide expansion to reduce upfront costs.

OPERATIONAL CHALLENGES

- **Training Programs:** Train artisans, guides, and homestay owners in digital tools.
- **Local Digital Promoters:** Appoint youth volunteers to help communities onboard the platform.
- **Incentives:** Provide verified vendors with higher visibility and promotional benefits.

Business Model

FUNDING SOURCES FOR SUSTAINABILITY



SUPPORTING SOURCES



TARGET CUSTOMERS

Tourists, local businesses and service providers, tribal artisans, event and community organizers, local administration, state tourism department, and cultural and tribal welfare departments.

FINANCIAL MODEL

- **Government Schemes:** Central/state tourism development funds. Digital India and Incredible India initiatives.
- **CSR & PPP Models:** Tech companies and telecoms as partners reduce financial burden.
- **Phased rollouts reduces upfront costs.**

VALUE ADDED SERVICES

- **Tourists:** Itinerary planner, emergency aid, verified services, AR/VR.
- **Community:** Microfinance, e-certifications, visibility, marketplace, event promos.
- **Government:** Tourist flow analysis, sustainability tracking, cross-department integration.

SCALABILITY

Start with one/two districts that are tourist hotspots
→ onboarding of the more digitally acquainted first
→ starting with digital literacy programs for local communities and service providers → onboarding local communities and partnerships with private businesses → state wide scaling

BENEFITS OF THE SOLUTION

Tourists

- **Personalized Experience:** AI creates itineraries based on budget and interests. 24/7 multilingual chatbot + real-time maps + recommendations + navigation.
- **Transparency & Trust:** Blockchain-verified guides, homestays ensure scam-free, safe, and reliable experiences.

For Local Communities

- **Economic Empowerment:** Artisans, guides, and homestays gain direct market access.
- **Fair Trade:** Blockchain ensures secure payments and verifications to uplift local communities.

For Government & Policymakers

- **Data-Driven Insights:** Feedback analysis helps identify gaps and boosts regional revenue (transport, food, lodging).
- **Efficient Governance:** Increased footfall allows real-time monitoring of tourist trends and service quality.

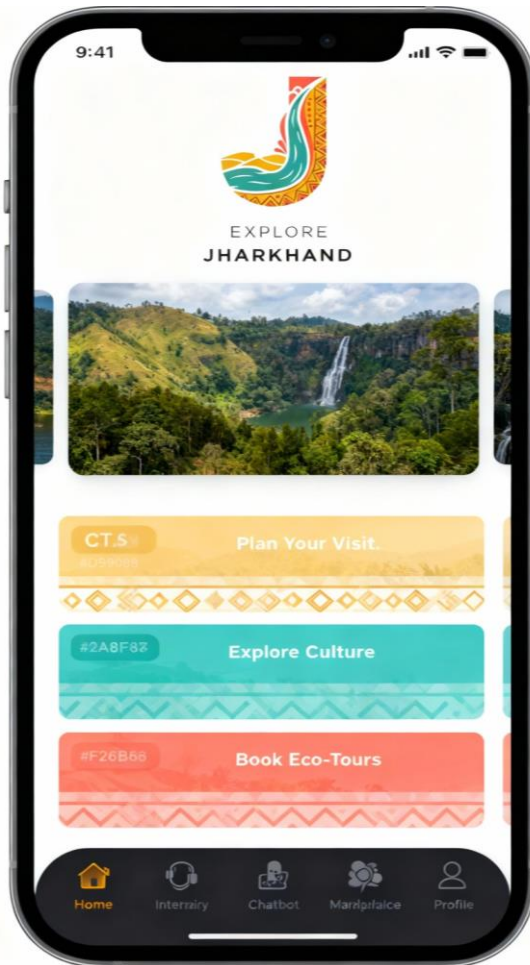
For The Environment

- **Sustainable Tourism:** Promotes eco-tourism and responsible travel practices.
- **Controlled Growth:** Smart itineraries encourage practices like homestays, agro-tourism, and low-impact travel.

Potential impacts on the target audience

- **For Economy:** Boosts tourist inflow, jobs, and state revenue.
- **For Culture:** Showcases tribal traditions, festivals, and crafts.
- **For Society:** Promotes inclusivity, enhances digital literacy, and encourages social openness.
- **For Infrastructure:** Creates a digital backbone that enables integration of future innovations.
- **For Governance:** AI-based data and feedback analysis support data-driven decision-making and improve the overall tourist experience.
- **For Tourism:** Footfall trends aid planning of events and promotions.

Sample UI development by the team



Reference App & Web



Reference Article & Sources

- Digital tourism platforms, sustainability, & Industry 4.0: [The Role of Digital Tourism Platforms in Advancing Sustainable Development Goals in the Industry 4.0 Era](#)
- Blockchain in tourism : [Blockchain technology diffusion in tourism: Evidence from ...](#)
- Digital transformation of tourism business models : [Digital Transformation of the Tourism Business in an Open Economy](#)
- Role of social media & digital outreach in tourism : [The role of social media content in attracting tourists](#)
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